

Experiment/Case Study No : 8

Aim:

Access and validate form fields using events; design a gym admission form with live input checks.

Software/Tools Required

- Visual Studio Code
- Google Chrome (Web Browser)

Experiment Program Code

1. Index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Gym Admission Form</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<div class="form-card">
<h1> 🏋️ Gym Admission</h1>
<p>Fill in your details to get started</p>
<div class="progress-bar"><div id="bar"></div></div>
<form id="gymForm" novalidate>

<div class="group" id="g-name">
<label>Full Name</label>
<input type="text" id="name" placeholder="e.g. Hussain Khan" autocomplete="off">
<span class="err-msg" id="e-name"></span>
<span class="ok-msg">✓ Looks good</span>
</div>

<div class="row">
<div class="group" id="g-age">
<label>Age</label>
<input type="number" id="age" placeholder="16-70" min="16" max="70">
<span class="err-msg" id="e-age"></span>
<span class="ok-msg">✓ Valid</span>
</div>

<div class="group" id="g-gender">
<label>Gender</label>
<select id="gender">
<option value="">Select</option>
<option>Male</option>
<option>Female</option>
<option>Other</option>
</select>
<span class="err-msg" id="e-gender"></span>
```

```

<span class="ok-msg">✓ Selected</span>
</div>

</div>
<div class="row">
<div class="group" id="g-phone">
<label>Phone</label>
<input type="tel" id="phone" placeholder="10-digit number" maxlength="10"
inputmode="numeric">
<span class="err-msg" id="e-phone"></span>
<span class="ok-msg">✓ Valid</span>
</div>

<div class="group" id="g-email">
<label>Email</label>
<input type="email" id="email" placeholder="you@example.com" autocomplete="off">
<span class="err-msg" id="e-email"></span>
<span class="ok-msg">✓ Valid</span>
</div>

</div>
<div class="group" id="g-plan">
<label>Membership Plan</label>
<select id="plan">
<option value="">Choose a plan</option>
<option value="monthly">Monthly - ₹2,499</option>
<option value="quarterly">Quarterly - ₹6,999</option>
<option value="yearly">Yearly - ₹19,999</option>
<option value="lifetime">Lifetime - ₹99,999</option>
</select>

<span class="err-msg" id="e-plan"></span>
<span class="ok-msg">✓ Selected</span>
</div>
<div class="group" id="g-address">
<label>Address</label>
<textarea id="address" rows="3" placeholder="House / Flat, Street, City, PIN"></textarea>
<span class="err-msg" id="e-address"></span>
<span class="ok-msg">✓ Filled</span>
</div>

<button type="submit" class="btn">Submit Admission</button>
</form>
</div>
<script src="script.js"></script>
</body>
</html>

```

2. style.css

```

*{margin:0;padding:0;box-sizing:border-box;font-family:'Segoe UI',sans-serif}
body{background:linear-gradient(135deg,#0f0c29,#302b63,#24243e);min-
height:100vh;display:flex;align-items:center;justify-content:center;padding:20px}
.form-card{background:#1a1a2e;border-radius:16px;padding:40px;width:100%;max-
width:520px;box-shadow:0 20px 60px rgba(0,0,0,.5)}
.form-card h1{text-align:center;color:#00e676;font-size:1.8rem;margin-bottom:8px}
.form-card p{text-align:center;color:#888;margin-bottom:28px;font-size:.9rem}
.group{margin-bottom:18px;position:relative}
.group label{display:block;color:#ccc;font-size:.85rem;margin-bottom:6px;font-weight:500}
.group input,.group select,.group textarea{width:100%;padding:12px 14px;border:2px solid
#333;border-radius:8px;background:#16213e;color:#fff;font-size:.95rem;transition:border-
color .3s,box-shadow .3s;outline:none}
.group input:focus,.group select:focus,.group textarea:focus{border-color:#00e676;box-

```

```

shadow:0 0 0 3px rgba(0,230,118,.15)}
.group.error input,.group.error select,.group.error textarea{border-color:#ff5252}
.group.success input,.group.success select,.group.success textarea{border-color:#00e676}
.err-msg{color:#ff5252;font-size:.78rem;margin-top:5px;display:none}
.ok-msg{color:#00e676;font-size:.78rem;margin-top:5px;display:none}
.group.error .err-msg{display:block}
.group.success .ok-msg{display:block}
.row{display:flex;gap:14px}
.row .group{flex:1}
.btn{width:100%;padding:14px;background:linear-
gradient(90deg,#00e676,#00b0ff);color:#000;border:none;border-radius:8px;font-
size:1rem;font-weight:700;cursor:pointer;margin-top:10px;transition:transform .2s,box-
shadow .2s}
.btn:hover{transform:translateY(-2px);box-shadow:0 8px 25px rgba(0,230,118,.3)}
.btn:active{transform:translateY(0)}
.progress-bar{height:4px;background:#333;border-radius:4px;margin-
bottom:24px;overflow:hidden}
.progress-bar div{height:100%;background:linear-
gradient(90deg,#00e676,#00b0ff);width:0%;transition:width .4s}

```

3. Script.js

```

const fields={

name:{el:document.getElementById('name'),group:document.getElementById('g-
name'),err:document.getElementById('e-name'),validate(v){v=v.trim();if(!v)return'Name is
required';if(v.length<3)return'At least 3 characters';if(!/^[A-Za-z\s\.'-]+$/.test(v))return'Letters,
spaces and . only';return''}},

age:{el:document.getElementById('age'),group:document.getElementById('g-
age'),err:document.getElementById('e-age'),validate(v){if(!v)return'Age is required';const
n=+v;if(n<16)return'Must be 16 or older';if(n>70)return'Max age is 70';return''}},

gender:{el:document.getElementById('gender'),group:document.getElementById('g-
gender'),err:document.getElementById('e-gender'),validate(v){return v?":Please select
gender"}},

phone:{el:document.getElementById('phone'),group:document.getElementById('g-
phone'),err:document.getElementById('e-phone'),validate(v){if(!v)return'Phone is
required';if(!/^\d{10}$/.test(v))return'Enter a valid 10-digit number';return''}},

email:{el:document.getElementById('email'),group:document.getElementById('g-
email'),err:document.getElementById('e-email'),validate(v){if(!v)return'Email is
required';if(!/^[^\s@]+@[^\s@]+\.[^\s@]{2,}$/.test(v))return'Enter a valid email';return''}},

plan:{el:document.getElementById('plan'),group:document.getElementById('g-
plan'),err:document.getElementById('e-plan'),validate(v){return v?":Please choose a plan"}},

address:{el:document.getElementById('address'),group:document.getElementById('g-
address'),err:document.getElementById('e-address'),validate(v){v=v.trim();if(!v)return'Address
is required';if(v.length<10)return'At least 10 characters';return''}}

};

const bar=document.getElementById('bar');

function check(key,showOk=true){

```

```

const f=fields[key];
const msg=f.validate(f.el.value);
f.group.classList.remove('error','success');
if(msg){f.group.classList.add('error');f.err.textContent=msg}
else if(showOk){f.group.classList.add('success')}
updateBar();
return !msg;
}

function updateBar(){
const total=Object.keys(fields).length;

const
done=Object.keys(fields).filter(k=>{f=fields[k];return!f.validate(f.el.value)&&f.el.value.trim()!=""}).length;

bar.style.width=(done/total*100)+'%';
}

Object.keys(fields).forEach(key=>{
const f=fields[key];
f.el.addEventListener('input',()=>check(key));
f.el.addEventListener('blur',()=>check(key));
});

document.getElementById('gymForm').addEventListener('submit',function(e){
e.preventDefault();
let allValid=true;

Object.keys(fields).forEach(k=>{if(!check(k))allValid=false});

if(allValid){
const name=fields.name.el.value.trim();

alert('🎉 Welcome, '+name+'! Your admission is confirmed.');
```

this.reset();

```

Object.keys(fields).forEach(k=>fields[k].group.classList.remove('success','error'));

bar.style.width='0%';
}

```

});
Output

< > C File C:/Users/MohammadHussain%20Khan/Music/M_Hussain_25070521505_a2/index.html



Gym Admision

Fill in your details to get started

Full Name

Hussain Khan

✓ Looks good

Age

20

✓ Valid

Gender

Male

✓ Selected

Phone

8482871166

✓ Valid

Email

you@example.com

Membership Plan

Choose a plan

Please choose a plan

Address

House / Flat, Street, City, PIN

Name: M HUSSAIN KHAN

PRN: 8482871166

Case Study Title

Design a Registration form for Gym Admission and use the appropriate the Form Validation and Events of JavaScript.

Case Study Program Code

1.index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Gym Admission Registration Form</title>
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link
href="https://fonts.googleapis.com/css2?family=Fredoka:wght@500;700&display=swa
p" rel="stylesheet">
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <div class="cartoon-card">
    <h2 class="form-title">GYM ADMISSION</h2>
    <form id="registrationForm">
      <table>
        <thead>
          <tr>
            <th>Field</th>
            <th>Details</th>
          </tr>
        </thead>
        <tbody>
          <tr>
            <td><label for="fullName">Name</label></td>
            <td>
              <input type="text" id="fullName" name="fullName"
placeholder="Your name...">
            </td>
          </tr>
          <tr>
```

```

<td>Sex</td>
<td>
  <label class="custom-radio">
    <input type="radio" id="male" name="sex" value="Male">
    <span>Male</span>
  </label>
  <label class="custom-radio">
    <input type="radio" id="female" name="sex" value="Female">
    <span>Female</span>
  </label>
</td>
</tr>
<tr>
<td><label for="eyeColor">Eye Color</label></td>
<td>
  <select id="eyeColor" name="eyeColor">
    <option value="">-- Select Color --</option>
    <option value="green">Green</option>
    <option value="blue">Blue</option>
    <option value="brown">Brown</option>
    <option value="black">Black</option>
  </select>
</td>
</tr>
<tr>
<td>Check All That Apply</td>
<td>
  <label class="custom-checkbox">
    <input type="checkbox" id="tall" name="attributes" value="Over 6
feet tall">
    <span>Over 6 feet tall</span>
  </label>
  <label class="custom-checkbox">
    <input type="checkbox" id="weight" name="attributes"
value="Over 200 pounds">
    <span>Over 200 pounds</span>
  </label>
</td>
</tr>
<tr>
<td colspan="2">
  <label for="ability" class="full-label">Describe your athletic
ability:</label>

```

```

        <textarea id="ability" name="ability" rows="4" placeholder="Tell us
about your fitness background..."></textarea>
    </td>
</tr>
<tr>
    <td colspan="2" class="button-cell">
        <button type="submit" class="cartoon-btn">Enter My
Information</button>
    </td>
</tr>
</tbody>
</table>
</form>
</div>
<div style="position: fixed; bottom: 10px; right: 10px; font-family: sans-serif;
font-size: 14px; color: #da1111; background-color: rgba(255, 208, 0, 0.8); padding:
5px 10px;
border-radius: 4px; z-index: 1000;">
    Name: M HUSSAIN KHAN <br>
    PRN: 8482871166
</div>

<script src="script.js"></script>
</body>
</html>

```

2. style.css

```

:root {
    --bg-color: #fef6e4;
    --card-bg: #ffffff;
    --primary-yellow: #ffcb05;
    --primary-cyan: #00e5ff;
    --primary-pink: #ff527b;
    --accent-purple: #8b5cf6;
    --border-color: #1e1e1e;
    --border-width: 3.5px;
    --shadow-distance: 6px;
}

* {
    box-sizing: border-box;
    font-family: 'Fredoka', cursive, sans-serif;
}

```



```
body {
  background-color: var(--bg-color);
  background-image: radial-gradient(var(--border-color) 12%, transparent 12%);
  background-size: 24px 24px;
  display: flex;
  justify-content: center;
  align-items: center;
  min-height: 100vh;
  margin: 0;
  padding: 20px;
}
```

```
.cartoon-card {
  background-color: var(--card-bg);
  border: var(--border-width) solid var(--border-color);
  border-radius: 20px;
  box-shadow: var(--shadow-distance) var(--shadow-distance) 0px var(--border-color);
  padding: 24px;
  max-width: 520px;
  width: 100%;
}
```

```
.form-title {
  margin: 0 0 16px 0;
  text-align: center;
  font-size: 28px;
  font-weight: 700;
  letter-spacing: 1px;
  color: var(--border-color);
  background-color: var(--primary-yellow);
  border: var(--border-width) solid var(--border-color);
  border-radius: 12px;
  padding: 10px;
  box-shadow: 4px 4px 0px var(--border-color);
  text-transform: uppercase;
}
```

```
table {
  width: 100%;
  border-collapse: separate;
```

```

border-spacing: 0;
border: var(--border-width) solid var(--border-color);
border-radius: 14px;
overflow: hidden;
background-color: #fff;
box-shadow: 4px 4px 0px var(--border-color);
}

th, td {
border-bottom: var(--border-width) solid var(--border-color);
border-right: var(--border-width) solid var(--border-color);
padding: 12px;
vertical-align: middle;
font-size: 15px;
color: var(--border-color);
}

th:last-child, td:last-child {
border-right: none;
}

tr:last-child td {
border-bottom: none;
}

th {
background-color: var(--primary-cyan);
font-size: 18px;
text-transform: uppercase;
letter-spacing: 0.5px;
}

td:first-child {
font-weight: 700;
background-color: #fdf2f8;
width: 40%;
}

input[type="text"], select, textarea {
width: 100%;
padding: 8px 12px;
border: var(--border-width) solid var(--border-color);

```

```
border-radius: 8px;
font-size: 14px;
background-color: #fff;
color: var(--border-color);
outline: none;
transition: transform 0.1s ease, box-shadow 0.1s ease;
}
```

```
input[type="text"]:focus, select:focus, textarea:focus {
  transform: translate(-2px, -2px);
  box-shadow: 3px 3px 0px var(--border-color);
  background-color: #fffde7;
}
```

```
textarea {
  resize: vertical;
  margin-top: 8px;
}
```

```
.full-label {
  display: block;
  margin-bottom: 4px;
}
```

```
.custom-radio, .custom-checkbox {
  display: flex;
  align-items: center;
  gap: 8px;
  margin-bottom: 6px;
  cursor: pointer;
  user-select: none;
  font-weight: 500;
}
```

```
.custom-radio:last-child, .custom-checkbox:last-child {
  margin-bottom: 0;
}
```

```
input[type="radio"], input[type="checkbox"] {
  width: 18px;
  height: 18px;
  accent-color: var(--primary-pink);
}
```

```

        cursor: pointer;
    }

.button-cell {
    text-align: center;
    background-color: #f3e8ff !important;
    padding: 16px;
}

.cartoon-btn {
    background-color: var(--primary-pink);
    color: #ffffff;
    font-size: 16px;
    font-weight: 700;
    text-transform: uppercase;
    border: var(--border-width) solid var(--border-color);
    border-radius: 12px;
    padding: 12px 24px;
    cursor: pointer;
    box-shadow: 4px 4px 0px var(--border-color);
    transition: all 0.1s ease;
    width: 100%;
}

.cartoon-btn:hover {
    background-color: #ff3366;
    transform: translate(-2px, -2px);
    box-shadow: 6px 6px 0px var(--border-color);
}

.cartoon-btn:active {
    transform: translate(2px, 2px);
    box-shadow: 2px 2px 0px var(--border-color);
}

```

3.script.js

```

document.getElementById("registrationForm").addEventListener("submit",
function(event) {
    event.preventDefault();

    let name = document.getElementById("fullName").value.trim();
    let sex = document.querySelector('input[name="sex"]:checked');

```

```
let eyeColor = document.getElementById("eyeColor").value;
let ability = document.getElementById("ability").value.trim();

if (name === "") {
    alert("Please enter your name.");
    return;
}

if (!sex) {
    alert("Please select your sex.");
    return;
}

if (eyeColor === "") {
    alert("Please select your eye color.");
    return;
}

if (ability === "") {
    alert("Please describe your athletic ability.");
    return;
}

alert("Form submitted successfully!");
this.submit();
});
```

Output

The screenshot shows a web browser window with a file path in the address bar. The main content is a 'GYM ADMISSION' form. The form has a yellow header with the title 'GYM ADMISSION'. Below the header is a table with two columns: 'FIELD' and 'DETAILS'. The table contains the following rows:

FIELD	DETAILS
Name	Your name...
Sex	☐ Male ☐ Female
Eye Color	-- Select Color --
Check All That Apply	☐ Over 6 feet tall ☐ Over 200 pounds
Describe your athletic ability: Tell us about your fitness background...	
ENTER MY INFORMATION	

In the bottom right corner of the browser window, there is a yellow box containing the following text:

Name: M HUSSAIN KHAN
PRN: 8482871166

Result/Conclusion

The program was executed successfully, and the required output was obtained.

